



An Online Paleoclimate Data Assimilation with a Deep Learning-based Network

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Introduction

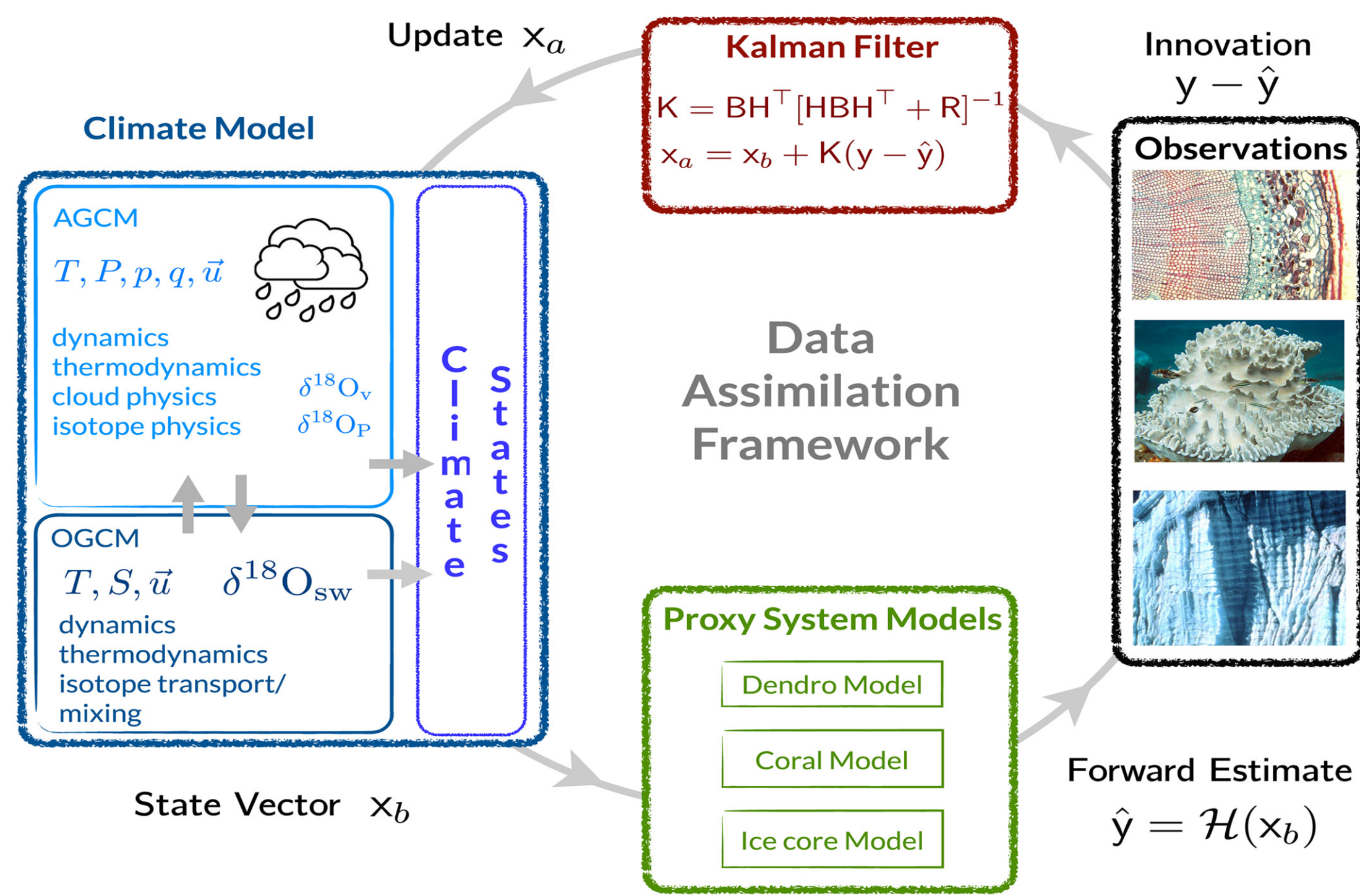


Figure 1. An ideal framework of paleoclimate data assimilation [1].

- Paleoclimate data assimilation combines proxies and climate models to produce an optimal estimate of past climate.
- Climate model is cost expensive! Ensemble integration of model for thousand years is impossible!!
- AI surrogate model is cost effective, but AI ensemble-based error covariances are inaccurate!

Surrogate models

- **Deep learning-based network:** A deep learning-based network (NET) is built to map the model state from year $t-1$ to year t . The NET is trained using output from ensemble simulations of climate models.

$$\mathbf{x}_t = M_{NET}(\mathbf{x}_{t-1}) \quad (1)$$

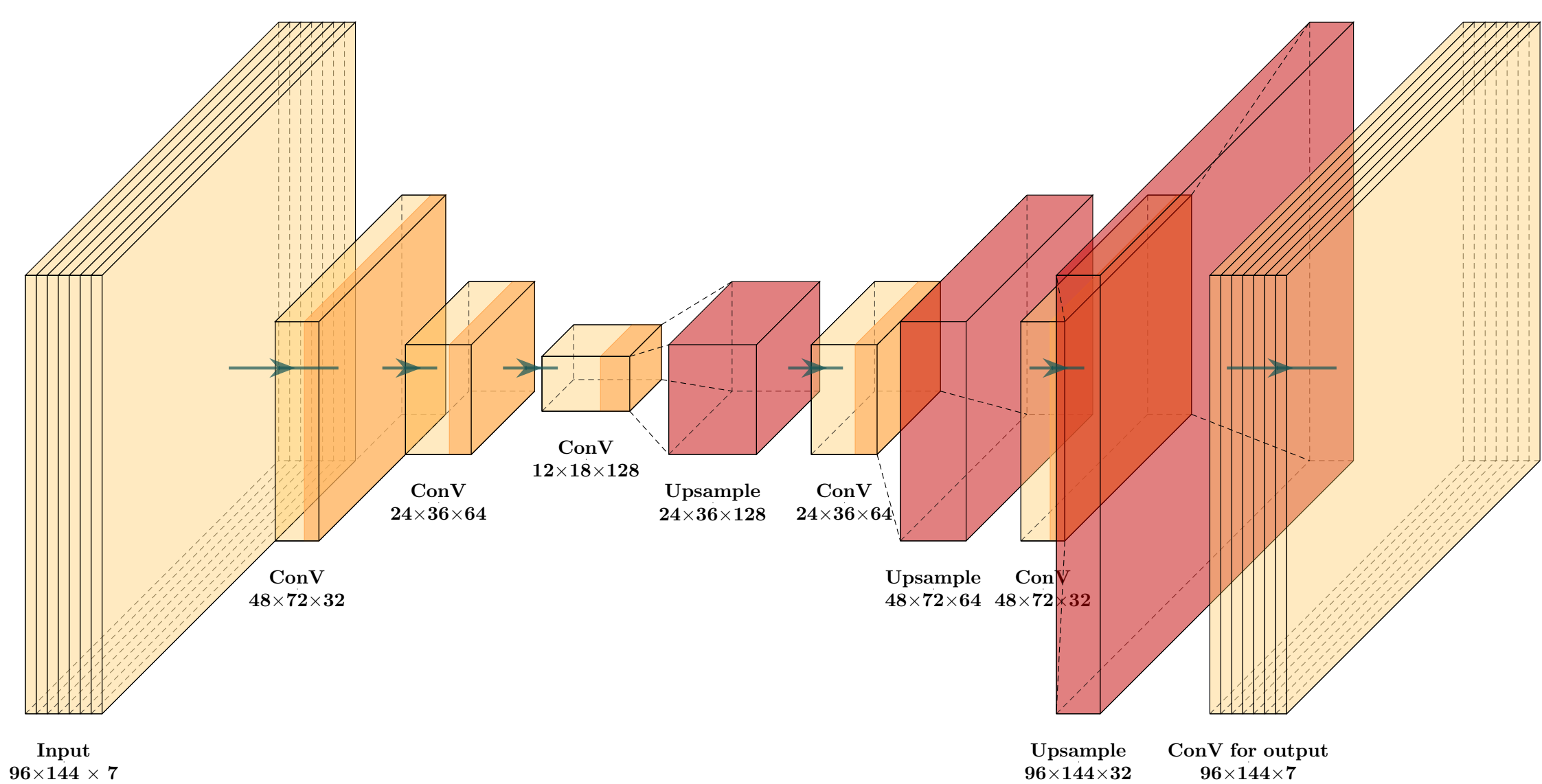


Figure 2. Architecture of the convolutional decoder-encoder network [2].

- **Linear inverse model as baseline:** A linear operator M_{LIM} that is estimated based on auto-regression.

Data assimilation methods

- **Ensemble square root filter (EnSRF):** The EnSRF uses ensemble priors and ensemble error covariances, which are modified in the following two ways:
- **Blend climate model-based ensemble:** The selected ensemble priors $\mathbf{x}_{climo}^f$ from climate models are added to the deep learning ensemble priors $\mathbf{x}_{cycle}^f$ with coefficient β :

$$\mathbf{x}_i^f = \beta \mathbf{x}_{cycle,i}^f + (1 - \beta) \mathbf{x}_{climo,i}^f \quad (2)$$

- **Hybrid climatological error covariances:** The hybrid background error covariances $\mathbf{P}_{hybrid}^f$ are a combination of the flow-dependent background error covariances $\mathbf{P}^f$ and the static background error covariances $\mathbf{P}_{climo}^f$, with weights α and $1 - \alpha$ respectively.

$$\mathbf{P}_{hybrid}^f = \alpha \mathbf{P}^f + (1 - \alpha) \mathbf{P}_{climo}^f \quad (3)$$

Experimental design

- **Proxy data:** The temperature sensitive proxy records from the PAGES 2k are assimilated, with temporal coverage of the Common Era (0-2000CE). The maximum number of proxies in 1945 CE is 554 (fig. 3).
- **Experiments:** Offline PDA (Exp_OFF), online PDA with LIM (Exp_LIM) and NET (Exp_NET) are conducted to reconstruct temperature.
- **Evaluations:** Root-mean square error (RMSE) and coefficient of efficiency (CE).

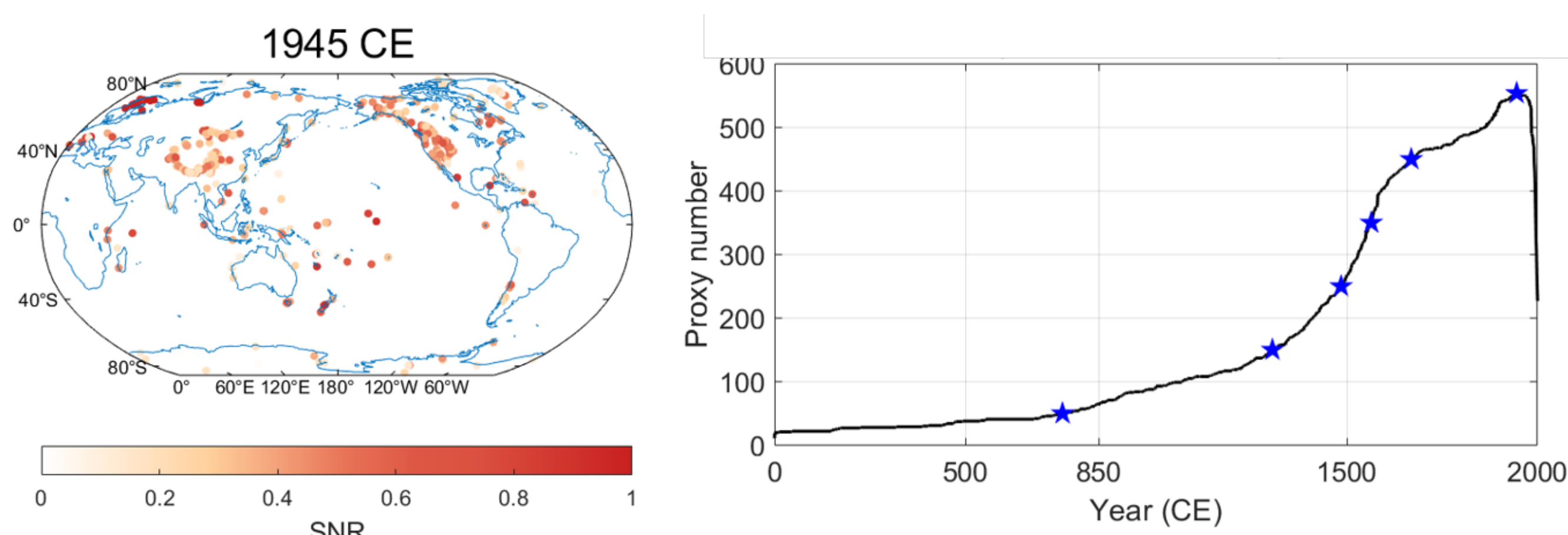


Figure 3. Temporal distributions of proxy records with their spatial distribution in 1945 CE

Predictive skills of surrogate models

- Compared to climatology, LIM has predictive skills for five years.
- Compared to LIM, NET further improves the predictive skills.

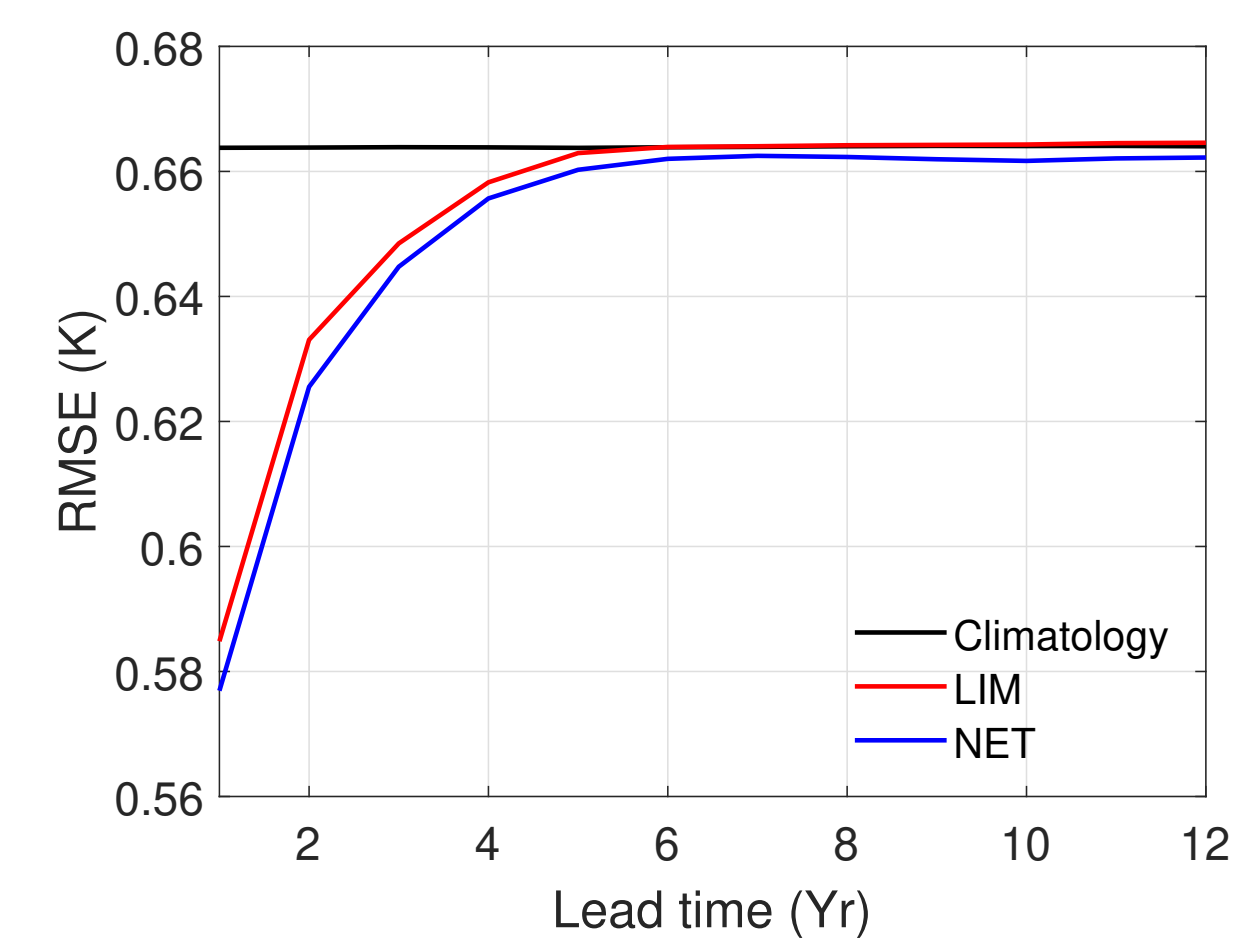


Figure 4. Globally averaged RMSEs at various forecast lead times (years)

Sensitivity experiments over the last 100 years

- Larger CE for Exp_LIM: Online PDA gets benefits from surrogate models than offline PDA.
- Largest CE for Exp_NET: Non-linearity obtained from AI model is more beneficial.
- Differences between Exp_LIM and Exp_NET are enlarged with sparse proxy data.

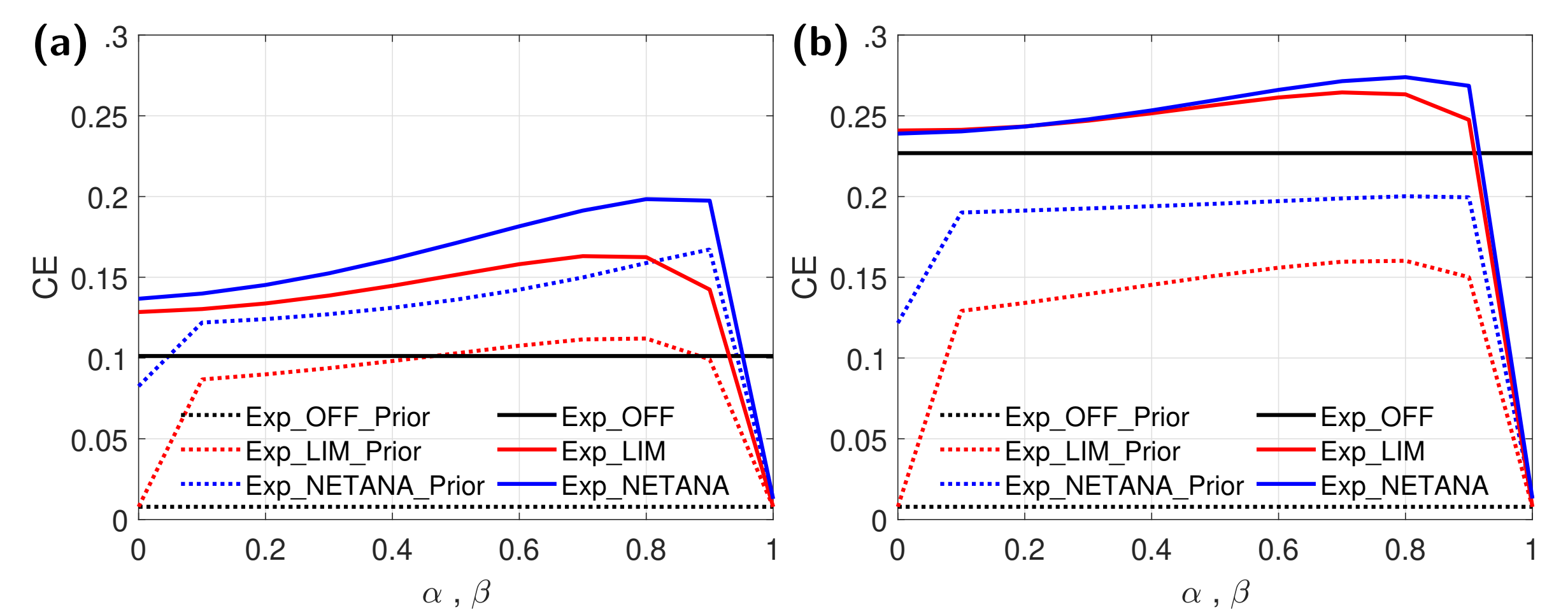


Figure 5. The last 100 years CE skills with (a) 150 (b) full pseudoproxies.

The last 2000 years reconstruction

- Online PDA produces larger variability with smaller uncertainty.

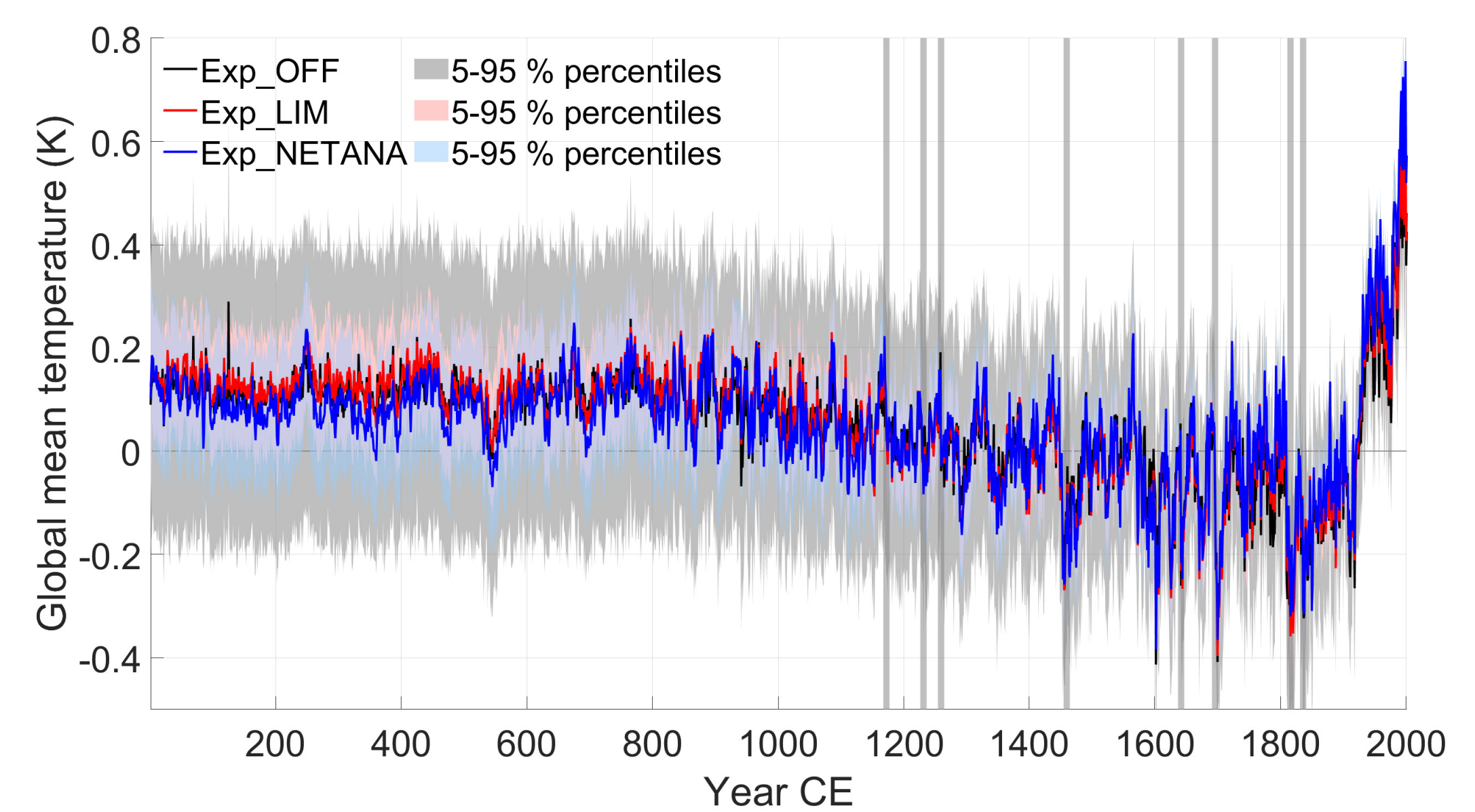


Figure 6. The last 2000 years reconstruction.

References

- [1] Gregory J. Hakim et al. "The last millennium climate reanalysis project: Framework and first results". In: *JGR Atmospheres* 121.12 (2016). DOI: 10.1002/2016JD024751.
- [2] Haohao Sun, Lili Lei, Zhengyu Liu, Liang Ning, and Zhe-Min Tan. "An Online Paleoclimate Data Assimilation With a Deep Learning-Based Network". In: *Journal of Advances in Modeling Earth Systems* 17.6 (2025), e2024MS004675. DOI: 10.1029/2024MS004675.